

2024 AMC 10B Problem 18 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 18

**Problem:**

How many different remainders can result when the 100th power of an integer is divided by 125 ?

Answer Choices:

- A. 1
- B. 2
- C. 5
- D. 25
- E. 125

Solution:

Write $N = 5k + r$ for $r = 0, 1, 2, 3$, or 4 . If $r = 0$, then $N = 5k$ and N^{100} is divisible by 125, so the remainder is 0. If $r = 1, 2, 3$, or 4 , then $N^2 = 25k^2 + 10rk + r^2 = 5m \pm 1$ for some integer m . Now use the Binomial Theorem:

$$\begin{aligned} N^{100} &= (N^2)^{50} = (5m \pm 1)^{50} = (5m)^{50} \pm 50(5m)^{49} + \binom{50}{2}(5m)^{48} \pm \dots \\ &\quad \pm \binom{50}{47}(5m)^3 + \binom{50}{48}(5m)^2 \pm 50(5m) + 1. \end{aligned}$$

All the terms except the final term have at least 3 factors of 5, so N^{100} has remainder 1 upon division by 125. Therefore there are only (B) 2 possible remainders: 0 and 1.

OR

Let $\phi(n)$ be the number of positive integers less than n that are relatively prime to n ; this is Euler's totient function. Then $\phi(125) = 5^3 - 5^2 = 100$. By Euler's Totient Theorem, if a is not a multiple of 5, then $a^{100} \equiv 1 \pmod{125}$. If a is a multiple of 5, then $a^{100} \equiv 0 \pmod{125}$. Therefore there are only (B) 2 possible remainders: 0 and 1.

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